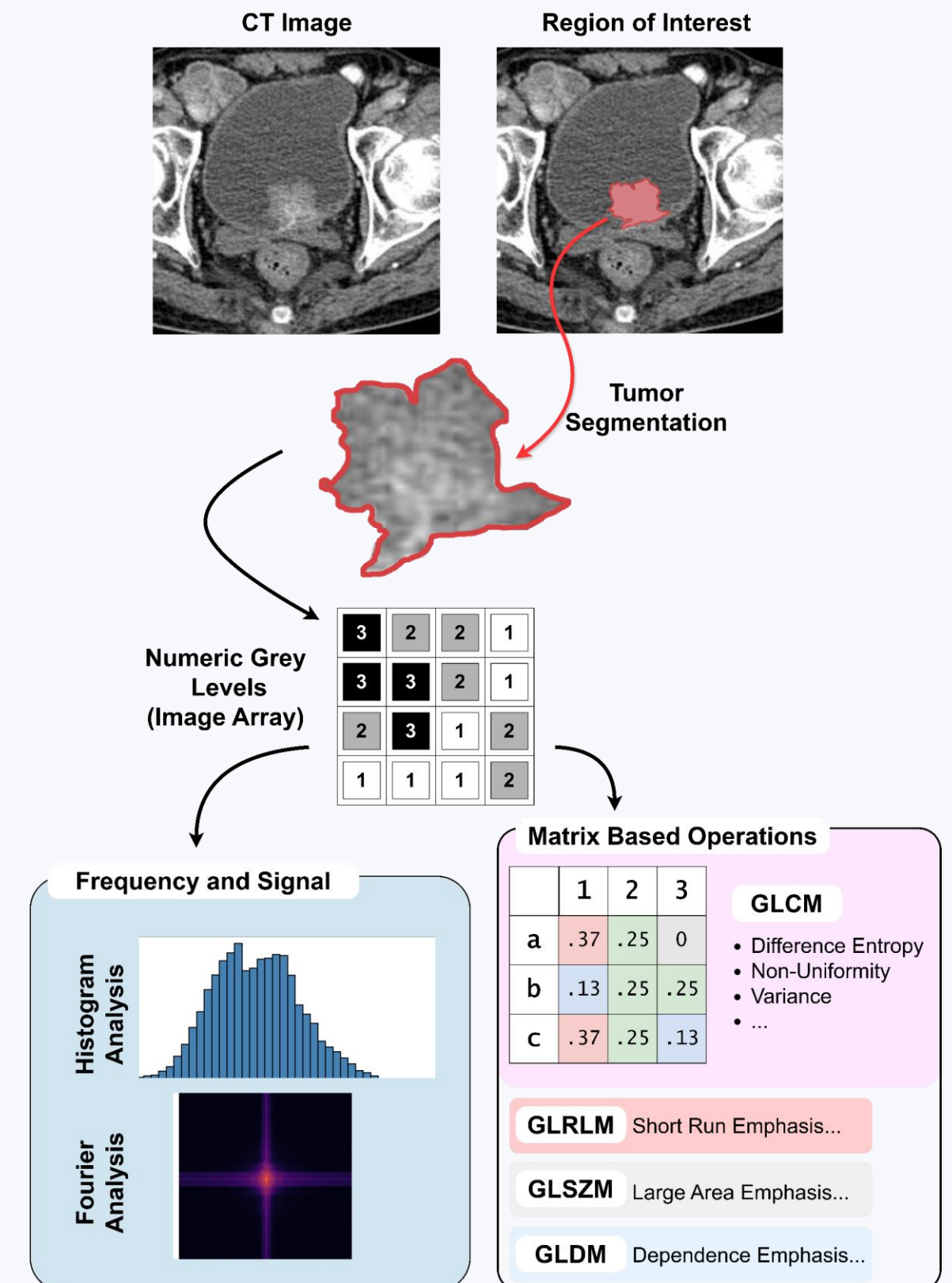


Background

- Genomic and transcriptomic profiling of muscle invasive bladder cancer (MIBC) has identified alterations that inform prognosis and treatment
- Radiomics – the extraction and analysis of quantitative imaging features – may provide a scalable and noninvasive way to link phenotypic imaging signatures to molecular alterations.

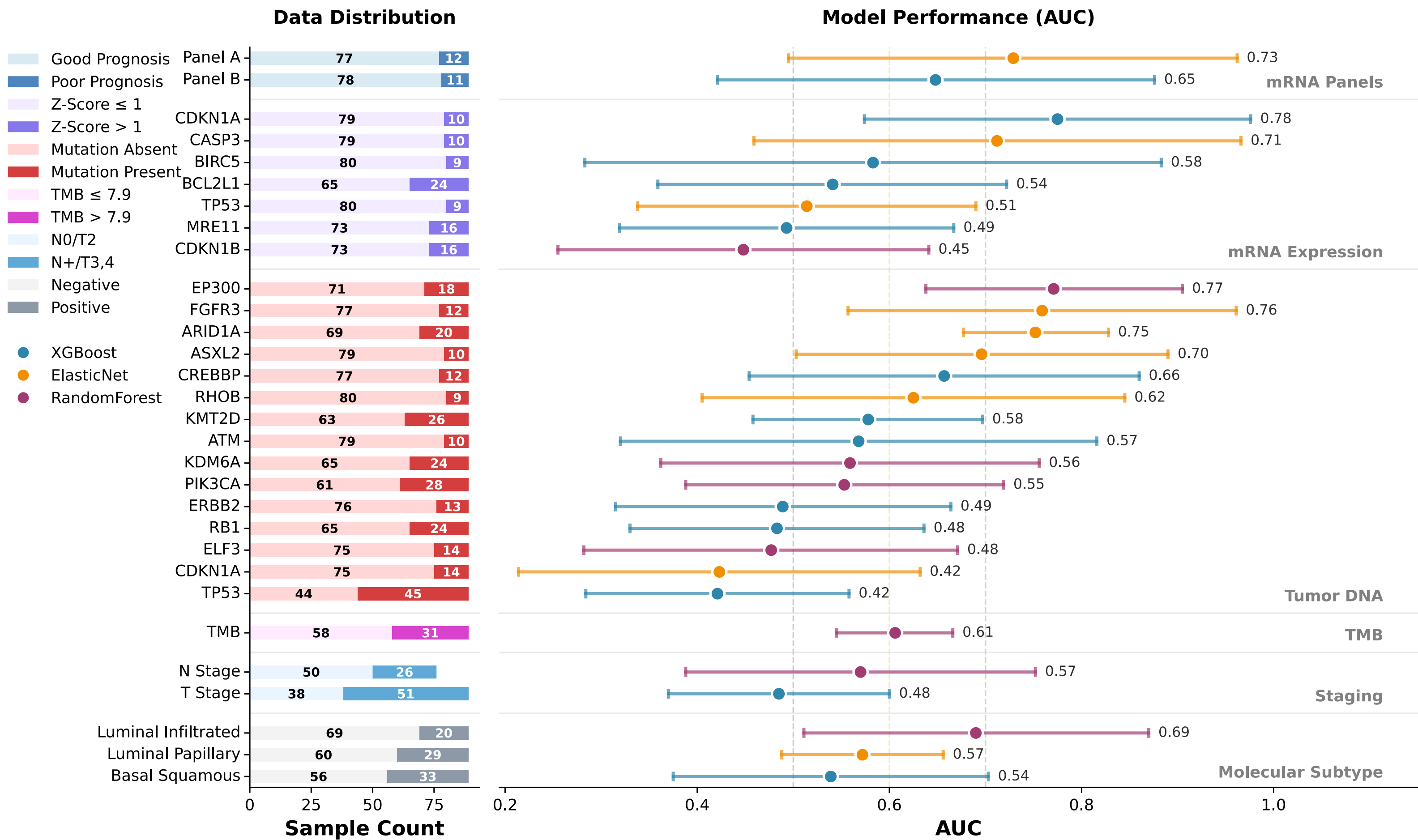
Methods

- We retrieved CT and genomic data from The Cancer Imaging Archive and The Cancer Genome atlas for 89 patients with MIBC
- Our radiomics pipeline extracted 488 CT image features used to train and evaluate 3 classifiers with a 10-fold cross validation

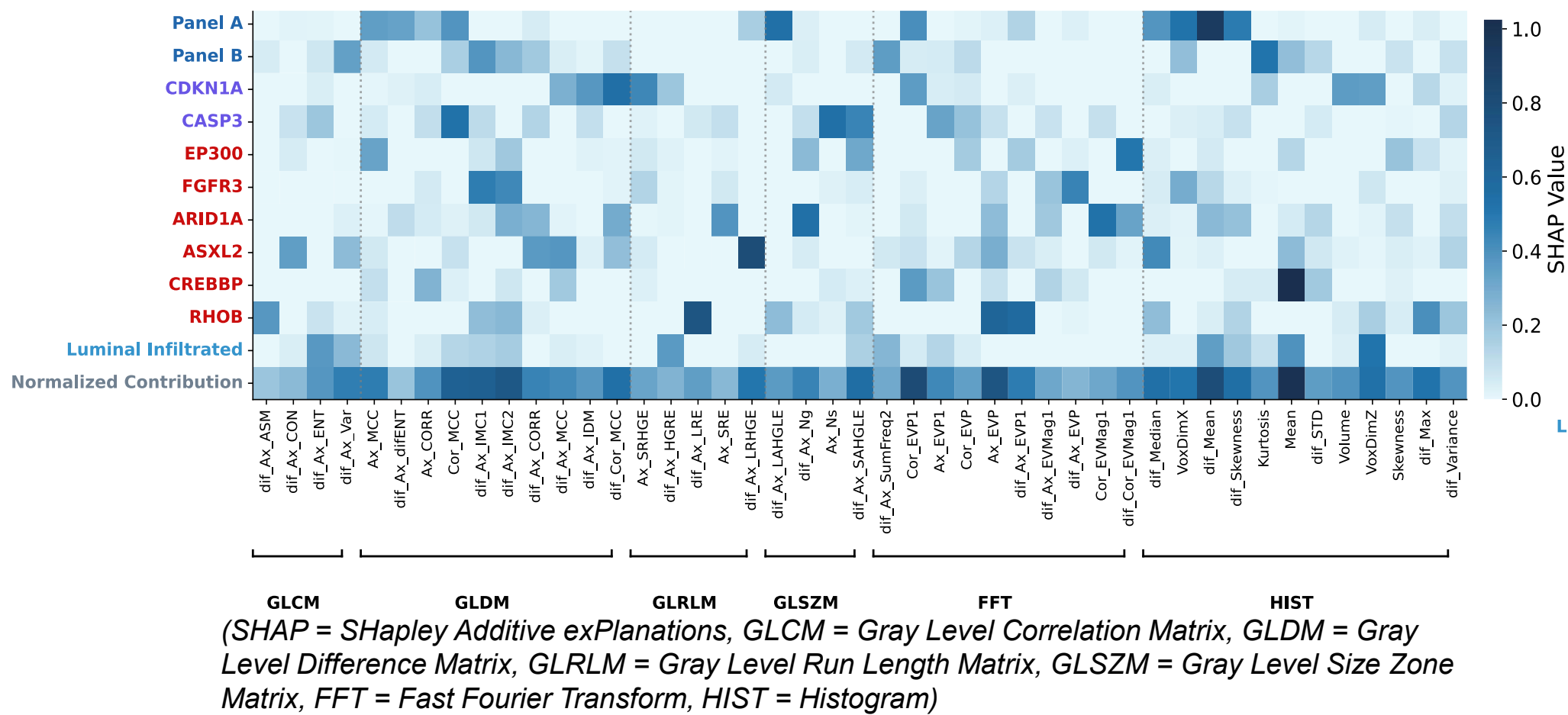


Results

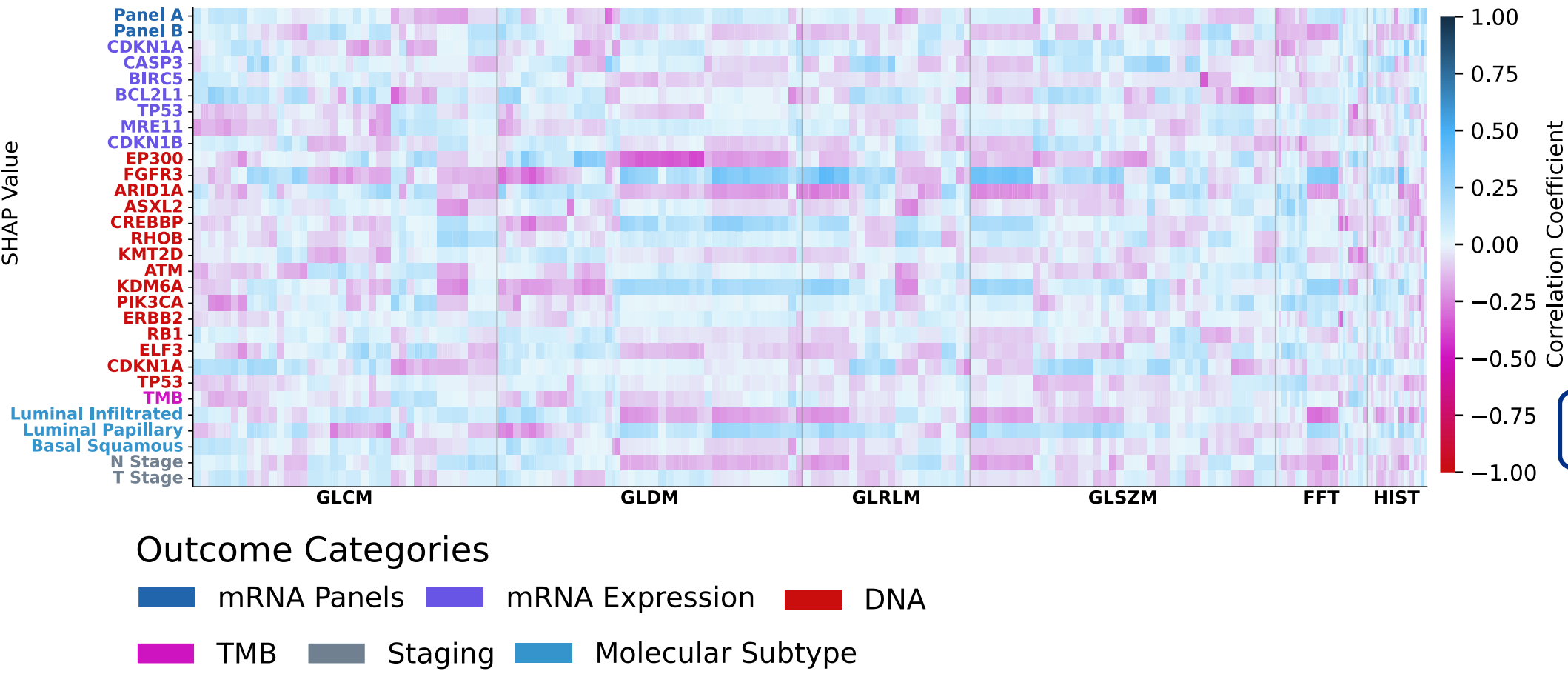
Radiomics Predicts Genomics & Transcriptomics Outcomes



Shapley Values for Most Influential Radiomics Variables



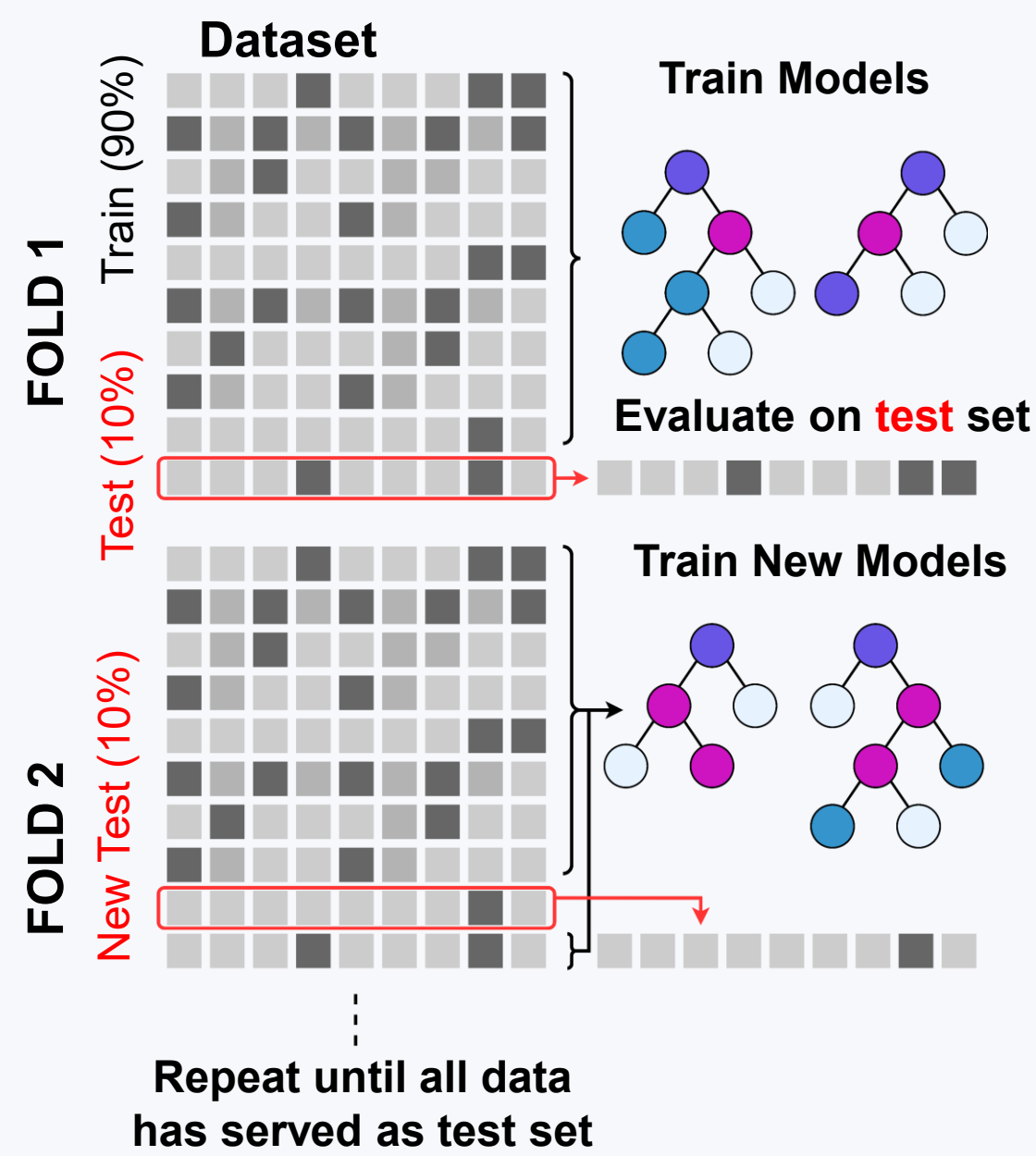
Point-Biserial Radiomic Feature and Outcome Correlations



Demographics

Characteristic	n	%
Gender		
Male	67	75%
Female	22	25%
Race		
White	79	89%
Black	9	10%
Asian	1	1%
Age (years ± SD)	69.3 ± 9.7	
Pathologic Tumor		
T2	38	43%
T3	40	45%
T4	11	12%
Metastasis Stage		
M0	43	48%
M1	3	3%
MX	43	49%
Lymph Node Stage		
N0	53	60%
N1	9	10%
N2	17	19%
NX	10	11%
Stage		
Stage II	31	35%
Stage III	31	35%
Stage IV	27	30%
mRNA cluster		
Basal squamous	33	37%
Luminal	5	6%
Luminal infiltrated	20	23%
Luminal papillary	29	33%
Neuronal	1	1%
Undetermined	1	1%
Total	89	100%

Cross Validation



Discussion & Limitations

- CT-based radiomics models can capture biologically and clinically relevant information in MIBC
- The majority (69%) of influential features were from the HIST, FFT, and GLDM feature classes
- This work demonstrates potential feasibility of noninvasive “virtual biopsy” to complement traditional tumor profiling
- Our models allow for a degree of interpretability not found in neural networks or transformer architectures
- Results are limited by the small dataset, poor prediction for several outcomes, and lack of external validation

Affiliations

- Scott Department of Urology, Dan L. Duncan Cancer Center, Baylor College of Medicine, Houston, TX
- Department of Radiology, Keck School of Medicine, 1500 San Pablo St, 2nd Floor, Los Angeles, CA 90033